

Computational Budget Report

GNR 638 Assignment 2

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1 Compute Environment

Final reported runs were executed on Kaggle with the following available resources:

- GPU: T4×2 (total available VRAM reported: 57.6 GB)
- System RAM: 30 GB
- Epoch time (user-reported): 30–60 seconds per epoch

2 Scope

This budget includes **only final reported runs** in the main report and excludes exploratory/aborted runs.

3 Run Accounting

Based on the final report configuration:

- Linear probe transfer: 3 models × 30 epochs = 90 epochs
- Fine-tuning strategies: 3 models × 4 strategies × 20 epochs = 240 epochs
- Few-shot learning: 3 models × 3 data regimes × 20 epochs = 180 epochs

Total training epochs counted:

$$E_{\text{total}} = 90 + 240 + 180 = 510 \text{ epochs}$$

4 Wall-Clock Estimate

Using the reported epoch-time range:

$$T_{\text{min}} = 510 \times 30\text{s} = 15,300\text{s} \approx 4.25 \text{ h}$$

$$T_{\text{max}} = 510 \times 60\text{s} = 30,600\text{s} \approx 8.50 \text{ h}$$

So the estimated training wall-clock for final runs is:

4.25 to 8.50 hours

5 Accelerator-Hour Estimate

If both T4 GPUs are reserved during these runs, accelerator-hours are:

$$\text{GPU-hours} = T_{\text{wall}} \times 2$$

Hence:

8.5 to 17.0 GPU-hours (T4-equivalent)

6 Experiment-wise Budget Split

Table 1: Estimated compute budget by experiment group (final runs only).

Experiment Group	Epochs	Wall-Clock (h)	T4 GPU-hours ($\times 2$)
Linear probe transfer	90	0.75–1.50	1.50–3.00
Fine-tuning strategies	240	2.00–4.00	4.00–8.00
Few-shot learning	180	1.50–3.00	3.00–6.00
Total	510	4.25–8.50	8.50–17.00

7 Notes and Assumptions

- Per-epoch time is modeled as a constant 30–60 s across all final runs, which may not reflect variability due to model size, batch size, or data regime.
- Layer-wise probing and corruption evaluation are primarily feature-extraction/inference workloads and are not included in the epoch-based training budget above.
- Kaggle usage may be free-tier for the user account; therefore direct monetary cost can be treated as zero for this report, while compute-equivalent cost is represented in GPU-hours.